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NUMERICAL ANALYSIS

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Numerical Analysis

4 Iterative Techniques for Solving Linear Systems

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4.1 Linear Fixed-Point Iteration

First we recall the definition of the eigenvalue of a matrix. The complex number $\lambda \in \mathbb{C}$ is an **eigenvalue** of the square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ if the linear system

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

has a nontrivial ($\mathbf{x} \neq \mathbf{0}$) solution. Its nontrivial solution $\mathbf{x}$ is called the **eigenvector** of the matrix $\mathbf{A}$ corresponding to the eigenvalue λ .

The eigenvector equation can be written in an equivalent form

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0},$$

where $\mathbf{I}$ denotes the $n \times n$ dimensional identity matrix, so $\mathbf{I}\mathbf{x} = \mathbf{x}$.

Theorem:

The $n \times n$ matrix $\mathbf{A}$ has n eigenvalues, which are solutions of the n th-degree algebraic equation

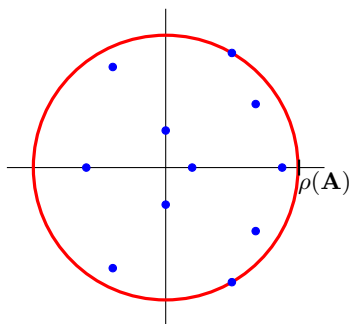
$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0,$$

*the so-called **characteristic equation**.*

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. The number

$$\rho(\mathbf{A}) := \max\{|\lambda| : \lambda \text{ is an eigenvalue of } \mathbf{A}\}$$

is called the **spectral radius** of $\mathbf{A}$.



Theorem:

Let $\|\cdot\|$ be a matrix norm. Then

$$\rho(\mathbf{A}) \leq \|\mathbf{A}\|.$$

Proof

Let λ be an eigenvalue of $\mathbf{A}$, and $\mathbf{v} \neq 0$ be a corresponding eigenvector, and let $\|\cdot\|$ be the vector norm which generates the matrix norm. Then

$$\lambda \mathbf{v} = \mathbf{A} \mathbf{v},$$

so

$$|\lambda| \|\mathbf{v}\| = \|\mathbf{A} \mathbf{v}\| \leq \|\mathbf{A}\| \|\mathbf{v}\|.$$

It follows

$$|\lambda| \leq \|\mathbf{A}\|,$$

therefore

$$\rho(\mathbf{A}) = \max |\lambda| \leq \|\mathbf{A}\|.$$

Next we investigate linear fixed-point equation

$$\mathbf{x} = \mathbf{T}\mathbf{x} + \mathbf{c},$$

where $\mathbf{T} \in \mathbb{R}^{n \times n}$, $\mathbf{c} \in \mathbb{R}^n$.

Consider the linear n dimensional fixed-point iteration

$$\mathbf{x}^{(k+1)} = \mathbf{T}\mathbf{x}^{(k)} + \mathbf{c}, \quad k = 0, 1, 2, \dots \quad (1)$$

First we consider the case when $\mathbf{c} = \mathbf{0}$. Then

$$\begin{aligned} \mathbf{x}^{(k)} &= \mathbf{T}\mathbf{x}^{(k-1)} \\ &= \mathbf{T}\left(\mathbf{T}\mathbf{x}^{(k-2)}\right) \\ &\vdots \\ &= \mathbf{T}^k \mathbf{x}^{(0)} \quad k = 1, 2, \dots \end{aligned}$$

For a real or complex constant a we have

$$\lim_{k \rightarrow \infty} a^k = 0 \quad \Longleftrightarrow \quad |a| < 1.$$

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Theorem:

Let $\mathbf{T} \in \mathbb{R}^{n \times n}$. The following statements are equivalent:

(i) $\lim_{k \rightarrow \infty} \mathbf{T}^k = \mathbf{0}$ (zero matrix), i.e.,

$$\lim_{k \rightarrow \infty} \|\mathbf{T}^k - \mathbf{0}\| = \lim_{k \rightarrow \infty} \|\mathbf{T}^k\| = 0$$

for any matrix norm $\|\cdot\|$;

(ii) $\lim_{k \rightarrow \infty} \mathbf{T}^k \mathbf{x} = \mathbf{0}$ (zero vector) for all $\mathbf{x} \in \mathbb{R}^n$, i.e.,

$$\lim_{k \rightarrow \infty} \|\mathbf{T}^k \mathbf{x} - \mathbf{0}\| = \lim_{k \rightarrow \infty} \|\mathbf{T}^k \mathbf{x}\| = 0$$

for all $\mathbf{x} \in \mathbb{R}^n$ and for any vector norm $\|\cdot\|$;

(iii) $\rho(\mathbf{T}) < 1$.

Proof

(i) $\Rightarrow$ (ii):

$$\|\mathbf{T}^k \mathbf{x}\| \leq \|\mathbf{T}^k\| \|\mathbf{x}\| \rightarrow 0, \quad \text{as } k \rightarrow \infty$$

for all $x \in \mathbb{R}^n$ and for any norm $\|\cdot\|$.

(ii) $\Rightarrow$ (iii):

Let λ be an eigenvalue of $\mathbf{T}$, and let $\mathbf{v}$ be an eigenvector corresponding to λ . Then

$$\mathbf{T}^k \mathbf{v} = \mathbf{T}^{k-1}(\mathbf{T} \mathbf{v}) = \lambda \mathbf{T}^{k-1} \mathbf{v} = \dots = \lambda^k \mathbf{v},$$

hence

$$\lim_{k \rightarrow \infty} \mathbf{T}^k \mathbf{v} = \mathbf{0} \quad \Rightarrow \quad |\lambda| < 1,$$

since $\mathbf{v} \neq \mathbf{0}$. Since λ was an arbitrary eigenvalue of $\mathbf{T}$,

$$\rho(\mathbf{T}) < 1$$

is satisfied.

(iii) $\Rightarrow$ (i): We do not prove it here.

Theorem:

If $\|\mathbf{T}\| < 1$ in some matrix norm $\|\cdot\|$, then $\|\mathbf{T}^k\| \rightarrow 0$ as $k \rightarrow \infty$.

Proof

The statement follows from

$$0 \leq \|\mathbf{T}^k\| \leq \|\mathbf{T}\|^k \rightarrow 0.$$

Theorem:

If $\|\mathbf{T}\| < 1$ in some matrix norm $\|\cdot\|$, then $\|\mathbf{T}^k\| \rightarrow 0$ as $k \rightarrow \infty$.

Proof

The statement follows from

$$0 \leq \|\mathbf{T}^k\| \leq \|\mathbf{T}\|^k \rightarrow 0.$$

For a real or complex number a we have that the infinite geometric sum

$$1 + a + a^2 + a^3 + \cdots + a^k + \cdots$$

is convergent if and only if $|a| < 1$, and

$$1 + a + a^2 + a^3 + \cdots = \frac{1}{1 - a}, \quad |a| < 1.$$

Next we investigate the convergence of the **geometric series** or **Neumann-series**

$$\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots ,$$

where $\mathbf{A}$ is a square matrix.

Theorem:

If $\rho(\mathbf{A}) < 1$, then the geometric series $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots$ is convergent, the matrix $\mathbf{I} - \mathbf{A}$ is invertible, and

$$(\mathbf{I} - \mathbf{A})^{-1} = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots .$$

Conversely, if the geometric series $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots$ is convergent, then

$$\rho(\mathbf{A}) < 1.$$

Proof

Let $\rho(\mathbf{A}) < 1$. Suppose that $\mathbf{I} - \mathbf{A}$ is not invertible. Then there exists a nonzero vector $\mathbf{x} \neq \mathbf{0}$ such that $(\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$. But then

$$\mathbf{A}\mathbf{x} = \mathbf{x},$$

i.e., 1 is an eigenvalue of $\mathbf{A}$, which contradicts to the assumption that $\rho(\mathbf{A}) < 1$. Hence $\mathbf{I} - \mathbf{A}$ is invertible.

It is easy to check that

$$(\mathbf{I} - \mathbf{A})(\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots + \mathbf{A}^m) = \mathbf{I} - \mathbf{A}^{m+1}. \quad (2)$$

Therefore,

$$\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots + \mathbf{A}^m = (\mathbf{I} - \mathbf{A})^{-1}(\mathbf{I} - \mathbf{A}^{m+1}),$$

and so, using that $\mathbf{A}^{m+1} \rightarrow \mathbf{0}$, we get

$$\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots + \mathbf{A}^m \rightarrow (\mathbf{I} - \mathbf{A})^{-1},$$

as $m \rightarrow \infty$.

Reverse statement is not proved here.

Corollary

If $\|\mathbf{A}\| < 1$ in some matrix norm $\|\cdot\|$, then the matrix $\mathbf{I} - \mathbf{A}$ is invertible, the geometric series $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \dots$ is convergent, $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \dots = (\mathbf{I} - \mathbf{A})^{-1}$, and

$$\|(\mathbf{I} - \mathbf{A})^{-1}\| \leq \frac{1}{1 - \|\mathbf{A}\|}.$$

Proof

$$\begin{aligned} \|(\mathbf{I} - \mathbf{A})^{-1}\| &= \left\| \lim_{m \rightarrow \infty} (\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \dots + \mathbf{A}^m) \right\| \\ &= \lim_{m \rightarrow \infty} \|\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \dots + \mathbf{A}^m\| \\ &\leq \lim_{m \rightarrow \infty} (\|\mathbf{I}\| + \|\mathbf{A}\| + \|\mathbf{A}^2\| + \|\mathbf{A}^3\| + \dots + \|\mathbf{A}^m\|) \\ &\leq \lim_{m \rightarrow \infty} (1 + \|\mathbf{A}\| + \|\mathbf{A}\|^2 + \|\mathbf{A}\|^3 + \dots + \|\mathbf{A}\|^m) \\ &= \frac{1}{1 - \|\mathbf{A}\|}. \end{aligned}$$

Theorem:

Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Let $\mathbf{A}$ be nonsingular, and

$$\|\mathbf{A} - \mathbf{B}\| < \frac{1}{\|\mathbf{A}^{-1}\|}.$$

Then $\mathbf{B}$ is also nonsingular, moreover,

$$\|\mathbf{B}^{-1}\| \leq \frac{\|\mathbf{A}^{-1}\|}{1 - \|\mathbf{A}^{-1}\|\|\mathbf{A} - \mathbf{B}\|}$$

and

$$\|\mathbf{A}^{-1} - \mathbf{B}^{-1}\| \leq \frac{\|\mathbf{A}^{-1}\|^2 \|\mathbf{A} - \mathbf{B}\|}{1 - \|\mathbf{A}^{-1}\|\|\mathbf{A} - \mathbf{B}\|}.$$

Proof

Consider the identities

$$\mathbf{B} = \mathbf{A} - (\mathbf{A} - \mathbf{B}) = \mathbf{A}(\mathbf{I} - \mathbf{A}^{-1}(\mathbf{A} - \mathbf{B})).$$

Using the assumption we get

$$\|\mathbf{A}^{-1}(\mathbf{A} - \mathbf{B})\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{A} - \mathbf{B}\| < 1,$$

therefore, the matrix

$$\mathbf{I} - \mathbf{A}^{-1}(\mathbf{A} - \mathbf{B})$$

is invertible. But then

$$\mathbf{B}^{-1} = (\mathbf{I} - \mathbf{A}^{-1}(\mathbf{A} - \mathbf{B}))^{-1} \mathbf{A}^{-1}$$

also exists. From this, relation

$$\mathbf{A}^{-1} - \mathbf{B}^{-1} = \mathbf{A}^{-1}(\mathbf{B} - \mathbf{A})\mathbf{B}^{-1}$$

and the last corollary imply the last two estimates of the theorem.

Consider again the fixed-point problem (1). The iteration yields

$$\begin{aligned}
 \mathbf{x}^{(k)} &= \mathbf{T}\mathbf{x}^{(k-1)} + \mathbf{c} \\
 &= \mathbf{T}(\mathbf{T}\mathbf{x}^{(k-2)} + \mathbf{c}) + \mathbf{c} \\
 &= \mathbf{T}^2\mathbf{x}^{(k-2)} + \mathbf{T}\mathbf{c} + \mathbf{c} \\
 &\vdots \\
 &= \mathbf{T}^k\mathbf{x}^{(0)} + (\mathbf{T}^{k-1} + \mathbf{T}^{k-2} + \cdots + \mathbf{T} + \mathbf{I})\mathbf{c}, \quad k = 1, 2, \dots
 \end{aligned}$$

Theorem:

Let $\mathbf{c} \neq \mathbf{0}$. The fixed-point equation

$$\mathbf{x} = \mathbf{T}\mathbf{x} + \mathbf{c}$$

has a unique solution, and the fixed-point iteration (1) converges to the unique solution of the equation for all $\mathbf{x}^{(0)} \in \mathbb{R}^n$ if and only if

$$\rho(\mathbf{T}) < 1.$$

Proof

Let $\rho(\mathbf{T}) < 1$. Then $\mathbf{I} - \mathbf{T}$ is invertible, hence the equation

$$\mathbf{x} = \mathbf{T}\mathbf{x} + \mathbf{c}$$

has a unique solution

$$\mathbf{x} = (\mathbf{I} - \mathbf{T})^{-1}\mathbf{c}.$$

We have

$$\mathbf{T}^k \mathbf{x}^{(0)} \rightarrow \mathbf{0}, \quad \text{for all } \mathbf{x}^{(0)} \in \mathbb{R}^n,$$

and

$$(\mathbf{T}^{k-1} + \mathbf{T}^{k-2} + \dots + \mathbf{T} + \mathbf{I})\mathbf{c} \rightarrow (\mathbf{I} - \mathbf{T})^{-1}\mathbf{c} \quad \text{as } k \rightarrow \infty,$$

therefore relation

$$\mathbf{x}^{(k)} = \mathbf{T}^k \mathbf{x}^{(0)} + (\mathbf{T}^{k-1} + \mathbf{T}^{k-2} + \dots + \mathbf{T} + \mathbf{I})\mathbf{c}$$

yields

$$\mathbf{x}^{(k)} \rightarrow \mathbf{x} \quad \text{as } k \rightarrow \infty.$$

Corollary

If $\|\mathbf{T}\| < 1$ in some matrix norm $\|\cdot\|$, then the iteration (1) is convergent for all initial value $\mathbf{x}^{(0)}$, and

$$\|\mathbf{x} - \mathbf{x}^{(k)}\| \leq \|\mathbf{T}\|^k \|\mathbf{x} - \mathbf{x}^{(0)}\|.$$

Proof

Subtracting the equations

$$\mathbf{x} = \mathbf{T}\mathbf{x} + \mathbf{c}$$

and

$$\mathbf{x}^{(k+1)} = \mathbf{T}\mathbf{x}^{(k)} + \mathbf{c}$$

we get

$$\mathbf{x} - \mathbf{x}^{(k+1)} = \mathbf{T}(\mathbf{x} - \mathbf{x}^{(k)}) = \dots = \mathbf{T}^{k+1}(\mathbf{x} - \mathbf{x}^{(0)}).$$

This yields the estimate or the statement.

Suppose that instead of the sequence

$$\mathbf{x}^{(k+1)} = \mathbf{T}\mathbf{x}^{(k)} + \mathbf{c}, \quad k = 0, 1, 2, \dots$$

we generate the sequence

$$\begin{aligned} \mathbf{y}^{(k+1)} &= \mathbf{T}\mathbf{y}^{(k)} + \mathbf{c} + \mathbf{w}^{(k+1)}, \quad k = 0, 1, \dots, \\ \mathbf{y}^{(0)} &= \mathbf{x}^{(0)} + \mathbf{w}^{(0)}, \end{aligned}$$

where the effect of the rounding error in the k th step is represented by $\mathbf{w}^{(k+1)}$, and $\mathbf{w}^{(0)}$ is the rounding error we get when we store the initial value of the sequence. We suppose that

$$\|\mathbf{w}^{(k)}\| \leq \varepsilon, \quad k = 0, 1, \dots$$

holds in a certain vector norm. We compute the difference of equations:

$$\mathbf{y}^{(k+1)} - \mathbf{x}^{(k+1)} = \mathbf{T}(\mathbf{y}^{(k)} - \mathbf{x}^{(k)}) + \mathbf{w}^{(k+1)}.$$

Then

$$\begin{aligned}
 \|\mathbf{y}^{(k+1)} - \mathbf{x}^{(k+1)}\| &\leq \|\mathbf{T}(\mathbf{y}^{(k)} - \mathbf{x}^{(k)})\| + \|\mathbf{w}^{(k+1)}\| \\
 &\leq \|\mathbf{T}\| \|\mathbf{y}^{(k)} - \mathbf{x}^{(k)}\| + \varepsilon \\
 &\vdots \\
 &\leq \|\mathbf{T}\|^{k+1} \|\mathbf{y}^{(0)} - \mathbf{x}^{(0)}\| + (\|\mathbf{T}\|^k + \dots + \|\mathbf{T}\| + 1)\varepsilon \\
 &\leq (\|\mathbf{T}\|^{k+1} + \|\mathbf{T}\|^k + \dots + \|\mathbf{T}\| + 1)\varepsilon.
 \end{aligned}$$

If $\|\mathbf{T}\| < 1$, then the last expression can be estimated by the sum of the geometric series:

$$\|\mathbf{y}^{(k+1)} - \mathbf{x}^{(k+1)}\| \leq \frac{1}{1 - \|\mathbf{T}\|} \varepsilon.$$

This shows that the computation is stable with respect to the rounding errors, and the smaller is $\|\mathbf{T}\|$ the smaller is the rounding error.

4.2 Jacobi Iteration

Example

Solve the linear system

$$\begin{aligned}4x_1 + 2x_2 - x_3 &= 9 \\5x_1 - 10x_2 + 2x_3 &= 8 \\-2x_1 + 3x_2 - 7x_3 &= 3.\end{aligned}$$

We rewrite the system in the equivalent form:

$$\begin{aligned}x_1 &= (9 - 2x_2 + x_3)/4 \\x_2 &= (-8 + 5x_1 + 2x_3)/10 \\x_3 &= (-3 - 2x_1 + 3x_2)/7.\end{aligned}$$

We define the sequences

$$\begin{aligned}x_1^{(k+1)} &= (9 - 2x_2^{(k)} + x_3^{(k)})/4 \\x_2^{(k+1)} &= (-8 + 5x_1^{(k)} + 2x_3^{(k)})/10 \\x_3^{(k+1)} &= (-3 - 2x_1^{(k)} + 3x_2^{(k)})/7\end{aligned}$$

for $k = 0, 1, 2, \dots$. This sequence is called **Jacobi iteration**.

Example cont.

Jacobi iteration

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0.0000000	0.0000000	0.0000000
1	2.2500000	-0.8000000	-0.4285714
2	2.5428571	0.2392857	-1.4142857
3	1.7767857	0.1885714	-1.0525510
4	1.8925765	-0.1221173	-0.8554082
$\vdots$	$\vdots$	$\vdots$	$\vdots$
19	2.0000268	-0.0000010	-1.0000107
20	1.9999978	0.0000112	-1.0000081
21	1.9999924	-0.0000027	-0.9999946
22	2.0000027	-0.0000027	-0.9999990
23	2.0000016	0.0000016	-1.0000019

Example cont.

The Jacobi iteration can be written in a vector form as

$$\mathbf{x}^{(k+1)} = \mathbf{T}\mathbf{x}^{(k)} + \mathbf{c},$$

where

$$\mathbf{T} = \begin{pmatrix} 0 & -2/4 & 1/4 \\ 5/10 & 0 & 2/10 \\ -2/7 & 3/7 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 9/4 \\ -8/10 \\ -3/7 \end{pmatrix}.$$

Since

$$\|\mathbf{T}\|_{\infty} = \max\{3/4, 7/10, 5/7\} = 3/4 < 1,$$

we get that the iteration is convergent.

Consider the linear system

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n = b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n = b_2 \\ \vdots & & \vdots & & & & \vdots \\ a_{n1}x_1 & + & a_{n2}x_2 & + & \dots & + & a_{nn}x_n = b_n. \end{array}$$

If $a_{ii} \neq 0$ for all $i = 1, \dots, n$, then the linear system can be transformed into the form

$$x_i = - \sum_{\substack{j=1 \\ j \neq i}}^n \frac{a_{ij}}{a_{ii}} x_j + \frac{b_i}{a_{ii}}, \quad i = 1, \dots, n,$$

and we can define the **Jacobi iteration** for $k = 0, 1, 2, \dots$ by

$$x_i^{(k+1)} = - \sum_{\substack{j=1 \\ j \neq i}}^n \frac{a_{ij}}{a_{ii}} x_j^{(k)} + \frac{b_i}{a_{ii}}, \quad i = 1, \dots, n. \quad (3)$$

If $a_{ii} = 0$ for some i , then we can try to interchange rows so that in the resulting matrix $a_{ii} \neq 0$ for all $i = 1, \dots, n$.

We introduce the notations $\mathbf{A} = \mathbf{L} + \mathbf{D} + \mathbf{U}$, where

$$\mathbf{L} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ a_{21} & 0 & 0 & \cdots & 0 \\ a_{31} & a_{32} & 0 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{n,n-1} & 0 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & 0 & a_{23} & \cdots & a_{2n} \\ 0 & 0 & 0 & \cdots & a_{3n} \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix},$$

and

$$\mathbf{D} = \text{diag}(a_{11}, a_{22}, \dots, a_{nn}).$$

$\mathbf{L}$ and $\mathbf{U}$ are lower and upper triangular matrices (with zeros in the diagonal too).

With this notation the linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

can be rewritten as

$$(\mathbf{L} + \mathbf{D} + \mathbf{U})\mathbf{x} = \mathbf{b}$$

hence

$$\mathbf{D}\mathbf{x} = -(\mathbf{L} + \mathbf{U})\mathbf{x} + \mathbf{b}.$$

Then multiplying this equation by $\mathbf{D}^{-1}$ we get a linear system of the form

$$\mathbf{x} = \mathbf{T}\mathbf{x} + \mathbf{c},$$

where

$$\mathbf{T} = \mathbf{T}_J := -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U}) \quad \text{and} \quad \mathbf{c} = \mathbf{D}^{-1}\mathbf{b}.$$

Therefore the Jacobi iteration can be defined as a fixed-point iteration

$$\mathbf{x}^{(k+1)} = \mathbf{T}\mathbf{x}^{(k)} + \mathbf{c}.$$

Theorem:

The Jacobi iteration is convergent for all initial values if and only if $\rho(\mathbf{T}_J) < 1$.

Corollary

If $\|\mathbf{T}_J\| < 1$ in some matrix norm $\|\cdot\|$, then the Jacobi iteration is convergent for all initial values $\mathbf{x}^{(0)}$.

In practice we can use the following sufficient condition.

Theorem:

If the matrix $\mathbf{A}$ is diagonally dominant, then the Jacobi iteration is convergent for all initial values $\mathbf{x}^{(0)}$.

Proof

Since

$$\mathbf{T}_J = \begin{pmatrix} 0 & -a_{12}/a_{11} & -a_{13}/a_{11} & \cdots & -a_{1n}/a_{11} \\ -a_{21}/a_{22} & 0 & -a_{23}/a_{22} & \cdots & -a_{2n}/a_{22} \\ -a_{31}/a_{33} & -a_{32}/a_{33} & 0 & \cdots & -a_{3n}/a_{33} \\ \vdots & & & \ddots & \vdots \\ -a_{n1}/a_{nn} & -a_{n2}/a_{nn} & -a_{n3}/a_{nn} & \cdots & 0 \end{pmatrix},$$

using the diagonal dominance of $\mathbf{A}$, we get

$$\|\mathbf{T}_J\|_\infty = \max_{i=1,\dots,n} \left\{ \sum_{\substack{j=1 \\ j \neq i}}^n \frac{|a_{ij}|}{|a_{ii}|} \right\} < 1.$$

Hence the Jacobi iteration is convergent.

4.3 Gauss–Seidel Iteration

Example

Consider again the system of the previous example, and its equivalent form

$$x_1 = (9 - 2x_2 + x_3)/4$$

$$x_2 = (-8 + 5x_1 + 2x_3)/10$$

$$x_3 = (-3 - 2x_1 + 3x_2)/7.$$

Define the iteration

$$x_1^{(k+1)} = (9 - 2x_2^{(k)} + x_3^{(k)})/4$$

$$x_2^{(k+1)} = (-8 + 5x_1^{(k+1)} + 2x_3^{(k)})/10$$

$$x_3^{(k+1)} = (-3 - 2x_1^{(k+1)} + 3x_2^{(k+1)})/7.$$

This is called **Gauss-Seidel iteration**.

Example cont.

Gauss–Seidel iteration

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0.0000000	0.0000000	0.0000000
1	2.2500000	0.3250000	-0.9321429
2	1.8544643	-0.0591964	-0.9837883
3	2.0336511	0.0200679	-1.0010141
4	1.9897125	-0.0053466	-0.9993521
5	2.0028353	0.0015472	-1.0001470
6	1.9991897	-0.0004346	-0.9999547
7	2.0002286	0.0001234	-1.0000124
8	1.9999352	-0.0000349	-0.9999964
9	2.0000183	0.0000099	-1.0000010
10	1.9999948	-0.0000028	-0.9999997
11	2.0000015	0.0000008	-1.0000001
12	1.9999996	-0.0000002	-1.0000000

We define the **Gauss–Seidel iteration** to solve the n -dimensional linear system. For $k = 0, 1, 2, \dots$ (if $a_{ii} \neq 0$ for all $i = 1, \dots, n$) we define

$$x_i^{(k+1)} = - \sum_{j=1}^{i-1} \frac{a_{ij}}{a_{ii}} x_j^{(k+1)} - \sum_{j=i+1}^n \frac{a_{ij}}{a_{ii}} x_j^{(k)} + \frac{b_i}{a_{ii}}, \quad i = 1, \dots, n. \quad (4)$$

Equation (4) can be rearranged to

$$\sum_{j=1}^i a_{ij} x_j^{(k+1)} = - \sum_{j=i+1}^n a_{ij} x_j^{(k)} + b_i, \quad i = 1, \dots, n,$$

i.e., using matrix notation,

$$(\mathbf{D} + \mathbf{L})\mathbf{x}^{(k+1)} = -\mathbf{U}\mathbf{x}^{(k)} + \mathbf{b},$$

where $\mathbf{L}$, $\mathbf{D}$, $\mathbf{U}$ is defined in the previous section. So the Gauss–Seidel iteration can be written in the form

$$\mathbf{x}^{(k+1)} = \mathbf{T}\mathbf{x}^{(k)} + \mathbf{c}$$

with

$$\mathbf{T} = \mathbf{T}_G := -(\mathbf{D} + \mathbf{L})^{-1}\mathbf{U} \quad \text{and} \quad \mathbf{c} = (\mathbf{D} + \mathbf{L})^{-1}\mathbf{b}.$$

Theorem:

The Gauss–Seidel iteration is convergent for any initial value if and only if $\rho(\mathbf{T}_G) < 1$.

Corollary

If $\|\mathbf{T}_G\| < 1$ in some matrix norm $\|\cdot\|$, then the Gauss–Seidel iteration is convergent for all initial values $\mathbf{x}^{(0)}$.

Theorem:

If $\mathbf{A}$ is diagonally dominant, then the Gauss–Seidel iteration is convergent for all initial values $\mathbf{x}^{(0)}$.

4.4 Error Bounds and the Condition Number

We can introduce stopping criteria for the Jacobi and the Gauss–Seidel iterations similar to nonlinear iterations:

$$(i) \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\| < \varepsilon, \quad (ii) \frac{\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|}{\|\mathbf{x}^{(k+1)}\|} < \varepsilon, \quad (iii) \|\mathbf{b} - \mathbf{A}\mathbf{x}^{(k)}\| < \varepsilon.$$

The vector

$$\mathbf{r} := \mathbf{b} - \mathbf{A}\tilde{\mathbf{x}}$$

is called the **residual vector** of the approximate solution $\tilde{\mathbf{x}}$ of the linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}.$$

Hypothesis: if the norm of $\mathbf{r}$ is small, then $\tilde{\mathbf{x}}$ is a good approximation of the exact solution $\mathbf{x}$.

Example

The exact solution of the linear system

$$\begin{pmatrix} 4 & 1 \\ 4.03 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 5.03 \end{pmatrix}$$

is $\mathbf{x} = (1, 1)^T$. Consider

$$\tilde{\mathbf{x}} = (2, -3)^T$$

as the “approximate” solution. The corresponding residual vector is

$$\mathbf{r} = \mathbf{b} - \mathbf{A}\tilde{\mathbf{x}} = (0, 0.03)^T.$$

Its infinity norm is

$$\|\mathbf{r}\|_{\infty} = 0.03,$$

which is small, but $\tilde{\mathbf{x}}$ cannot be considered as a good approximation of the true solution.

The next result gives conditions which imply that the smallness of the norm of $\|\mathbf{r}\|$ yields that the error of the approximation is also small.

Theorem:

Let $\mathbf{A}$ be a nonsingular square matrix, $\mathbf{x}$ be the exact solution of the system

$$\mathbf{Ax} = \mathbf{b},$$

the vector $\tilde{\mathbf{x}}$ is an approximate solution, and

$$\mathbf{r} := \mathbf{b} - \mathbf{A}\tilde{\mathbf{x}}.$$

Then

$$\|\mathbf{x} - \tilde{\mathbf{x}}\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{r}\|,$$

and

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| \|\mathbf{A}^{-1}\| \frac{\|\mathbf{r}\|}{\|\mathbf{b}\|}.$$

Proof

Subtracting relations

$$\mathbf{A}\mathbf{x} = \mathbf{b},$$

$$\mathbf{A}\tilde{\mathbf{x}} = \mathbf{b} - \mathbf{r}$$

we get

$$\mathbf{A}(\mathbf{x} - \tilde{\mathbf{x}}) = \mathbf{r},$$

and hence

$$\mathbf{x} - \tilde{\mathbf{x}} = \mathbf{A}^{-1}\mathbf{r}.$$

Then

$$\|\mathbf{x} - \tilde{\mathbf{x}}\| = \|\mathbf{A}^{-1}\mathbf{r}\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{r}\|.$$

Combining it with

$$\|\mathbf{b}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$$

we get

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}\| \|\mathbf{A}^{-1}\| \|\mathbf{r}\|}{\|\mathbf{A}\| \|\mathbf{x}\|} \leq \|\mathbf{A}\| \|\mathbf{A}^{-1}\| \frac{\|\mathbf{r}\|}{\|\mathbf{b}\|}.$$

The number

$$\text{cond}(\mathbf{A}) := \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$$

is called the **condition number** of the matrix $\mathbf{A}$ relative to a norm $\|\cdot\|$.

The condition number corresponding to the $\|\cdot\|_p$ norm is denoted by $\text{cond}_p(\mathbf{A})$. If a condition number of the matrix $\mathbf{A}$ is “big”, then it is called **ill-conditioned**, otherwise it is called **well-conditioned**.

Example

Consider the coefficient matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 4.03 & 1 \end{pmatrix}$ of an earlier example.

We can check that

$$\mathbf{A}^{-1} = \begin{pmatrix} -33.33 & 33.33 \\ 143.3 & -133.3 \end{pmatrix},$$

and so $\|\mathbf{A}\|_\infty = 5.03$, $\|\mathbf{A}^{-1}\|_\infty = 267.6$. Therefore $\text{cond}_\infty(\mathbf{A}) = 1346$, hence $\mathbf{A}$ is ill-conditioned, and so $(2, -3)^T$ is not a good approximation of the true solution despite the fact that $\mathbf{r}$ is small in norm.

4.5 Perturbation of Linear Systems

Example

Consider the linear system

$$\begin{aligned}x_1 + \frac{1}{2}x_2 + \frac{1}{3}x_3 &= 1 \\ \frac{1}{2}x_1 + \frac{1}{3}x_2 + \frac{1}{4}x_3 &= 1 \\ \frac{1}{3}x_1 + \frac{1}{4}x_2 + \frac{1}{5}x_3 &= 1\end{aligned}$$

Its exact solution is $x_1 = 3$, $x_2 = -24$ and $x_3 = 30$. Next suppose we compute the coefficients of the previous linear system up to 3 decimal digits, i.e., consider the system

$$\begin{aligned}y_1 + 0.5y_2 + 0.333y_3 &= 1 \\ 0.5y_1 + 0.333y_2 + 0.25y_3 &= 1 \\ 0.333y_1 + 0.25y_2 + 0.2y_3 &= 1\end{aligned}$$

Its solution is $y_1 = 3.4460555$, $y_2 = -26.2735192$ and $y_3 = 32.1042167$. The difference of the two solutions is:

$$\begin{aligned}|x_1 - y_1| &= 0.4460555, & |x_2 - y_2| &= 2.2735192, & |x_3 - y_3| &= 2.1042167 \\ \frac{|x_1 - y_1|}{|x_1|} &= 0.1486852 & \frac{|x_2 - y_2|}{|x_2|} &= 0.09472997, & \frac{|x_3 - y_3|}{|x_3|} &= 0.07014056\end{aligned}$$

Consider the linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}. \quad (5)$$

Suppose that instead of (5) we solve the linear system

$$\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}, \quad (6)$$

where $\tilde{\mathbf{b}} := \mathbf{b} + \Delta\mathbf{b}$ is a perturbation of $\mathbf{b}$ by $\Delta\mathbf{b}$. Its exact solution is denoted by $\tilde{\mathbf{x}}$.

Theorem:

Let $\mathbf{A}$ be a nonsingular square matrix, $\mathbf{x}$ and $\tilde{\mathbf{x}}$ be solutions of the linear systems (5) and (6), respectively. Then

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \text{cond}(\mathbf{A}) \frac{\|\mathbf{b} - \tilde{\mathbf{b}}\|}{\|\mathbf{b}\|}.$$

Proof

Subtracting (5) and (6) we get we get

$$\mathbf{A}(\mathbf{x} - \tilde{\mathbf{x}}) = \mathbf{b} - \tilde{\mathbf{b}},$$

hence

$$\mathbf{x} - \tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{b} - \tilde{\mathbf{b}}),$$

therefore,

$$\|\mathbf{x} - \tilde{\mathbf{x}}\| = \|\mathbf{A}^{-1}(\mathbf{b} - \tilde{\mathbf{b}})\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{b} - \tilde{\mathbf{b}}\|.$$

Using this and the inequality

$$\|\mathbf{b}\| = \|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$$

it follows

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}\| \|\mathbf{A}^{-1}\| \|\mathbf{b} - \tilde{\mathbf{b}}\|}{\|\mathbf{A}\| \|\mathbf{x}\|} \leq \text{cond}(\mathbf{A}) \frac{\|\mathbf{b} - \tilde{\mathbf{b}}\|}{\|\mathbf{b}\|}.$$

Now we consider the general case, when we perturb both the coefficient matrix and the right-hand-side of the system. We consider the linear system

$$\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}, \quad (7)$$

where $\|\mathbf{b} - \tilde{\mathbf{b}}\|$ and $\|\mathbf{A} - \tilde{\mathbf{A}}\|$ are “small”.

Theorem:

Let $\mathbf{A}$ be a nonsingular square matrix, and $\tilde{\mathbf{A}}$ be such that

$$\|\mathbf{A} - \tilde{\mathbf{A}}\| < \frac{1}{\|\mathbf{A}^{-1}\|}.$$

Let $\mathbf{x}$ and $\tilde{\mathbf{x}}$ be the exact solutions of (5) and (7), respectively. Then

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond}(\mathbf{A})}{1 - \text{cond}(\mathbf{A}) \frac{\|\mathbf{A} - \tilde{\mathbf{A}}\|}{\|\mathbf{A}\|}} \left(\frac{\|\mathbf{A} - \tilde{\mathbf{A}}\|}{\|\mathbf{A}\|} + \frac{\|\mathbf{b} - \tilde{\mathbf{b}}\|}{\|\mathbf{b}\|} \right).$$

The following properties of the condition number can be proved easily.

Theorem:

Let $\|\cdot\|$ be a fixed matrix norm and $\text{cond}(\cdot)$ be the corresponding condition number function. Then

- (i) $\text{cond}(\mathbf{A}) \geq 1$,
- (ii) $\rho(\mathbf{A})\rho(\mathbf{A}^{-1}) \leq \text{cond}(\mathbf{A})$

hold for all invertible matrix $\mathbf{A}$.

The number $\text{cond}_*(\mathbf{A}) := \rho(\mathbf{A})\rho(\mathbf{A}^{-1})$ is called the **spectral condition number** of the matrix $\mathbf{A}$.

Theorem (Gastinel):

Let $\|\cdot\|$ be a matrix norm, $\mathbf{A}$ be invertible. Then

$$\frac{1}{\text{cond}(\mathbf{A})} = \min \left\{ \frac{\|\mathbf{A} - \mathbf{B}\|}{\|\mathbf{A}\|} : \mathbf{B} \text{ is singular} \right\}.$$

An example for an ill-conditioned matrix is the so-called **Hilbert-matrix**:

$$\mathbf{H}_n = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots & \frac{1}{n+2} \\ \vdots & & & & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n-1} \end{pmatrix}.$$

Spectral condition number of the Hilbert-matrix

n	$\text{cond}_*(\mathbf{H}_n)$	n	$\text{cond}_*(\mathbf{H}_n)$
3	$5.24 \cdot 10^2$	7	$7.45 \cdot 10^8$
4	$1.55 \cdot 10^4$	8	$1.53 \cdot 10^{10}$
5	$4.77 \cdot 10^5$	9	$4.93 \cdot 10^{11}$
6	$1.50 \cdot 10^6$	10	$1.60 \cdot 10^{13}$